Schrödinger Bridges for Discrete Diffusion Models

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Abstract

We present a comprehensive framework for Discrete Schrödinger Bridges (DSB) to enable flexible generative modeling and translation between arbitrary discrete distributions. Standard discrete diffusion models are constrained to a pure-noise initialization, limiting controllability for tasks that require structured source priors (Lou et al., 2024; Liu et al., 2022). Our approach generalizes diffusion by finding the optimal stochastic process connecting a discrete source distribution μ_0 and a target μ_N (De Bortoli et al., 2021; Shi et al., 2023). In this report, we establish the theoretical connections between Optimal Transport, discrete diffusion, and Schrödinger Bridges (Santambrogio, 2015). We address the computational intractability of ideal I-projections in path space by substituting them with M-projections (Maximum Likelihood Estimation) for local transition fitting. To mitigate the loss of explicit endpoint marginal matching inherent to M-projections, we introduce a novel Adversarial Marginal Matching technique using GAN-based discriminators (Kholkin et al., 2024; Nowozin et al., 2016). We demonstrate the efficacy of this unified framework through successful, smooth domain translation between MNIST handwritten digits (LeCun et al., 1998) and EMNIST handwritten letters (Cohen et al., 2017).

1 Introduction

Diffusion models have achieved unprecedented success in generative tasks, particularly for continuous modalities such as images and audio (Liu et al., 2022; Sahoo et al., 2025). However, standard diffusion is fundamentally formulated as a transport process from a pure noise distribution to a data distribution, which limits controllability for tasks that require structured source priors, especially in discrete settings (Lou et al., 2024).

To address this limitation, we utilize the Schrödinger Bridge (SB) framework (De Bortoli et al., 2021; Shi et al., 2023). A Schrödinger Bridge generalizes standard diffusion by finding the most likely stochastic process connecting a source distribution μ_0 and a target distribution μ_N, given a reference dynamic.

While common architectures like CycleGAN are often used for domain translation, Schrödinger Bridges offer fundamental advantages. CycleGAN utilizes deterministic generators, which are prone to mode collapse and only map endpoints. In contrast, Schrödinger Bridges define a stochastic transport process with meaningful intermediate trajectory states, achieving better mode coverage and offering control over the path itself (Liu et al., 2022). This work aims to build a mathematically grounded and practically trainable discrete SB framework.

2 Theoretical Background

2.1 Optimal Transport: Static to Entropy-Regularized

The foundation of the Schrödinger Bridge lies in Optimal Transport (Santambrogio, 2015). The classic goal of Optimal Transport is to move mass from a source distribution μ to a target ν at a minimal cost $c(x, y)$. The original Monge formulation seeks a deterministic map T such that pushing μ through T yields ν ($T_{\#}\mu = \nu$):

$$\min_{T:\, T_{\#}\mu=\nu} \int c\big(x, T(x)\big)\, d\mu(x) \tag{1}$$

This problem is often ill-posed and highly non-convex, and a deterministic map may fail to exist. For instance, if the source is a single Dirac mass $\mu = \delta_{x_0}$ while the target is a mixture of two Dirac masses $\nu = \frac{1}{2}\delta_{y_1} + \frac{1}{2}\delta_{y_2}$ with $y_1 \neq y_2$, then any deterministic map T must send x_0 to a single point, so $T_\sharp\mu = \delta_{T(x_0)}$ cannot equal ν; hence the Monge constraint $T_\sharp\mu = \nu$ is infeasible in this case. The Kantorovich relaxation resolves this by seeking a probabilistic transport plan (coupling) π:

$$\min_{\pi\in\Pi(\mu,\nu)} \int c(x,y)\, d\pi(x,y), \qquad \Pi(\mu,\nu) = \{\pi : \pi_x = \mu,\ \pi_y = \nu\} \tag{2}$$

where the set $\Pi(\mu,\nu)$ contains all joint distributions with marginals μ and ν. While convex, exact Optimal Transport can be brittle and expensive to compute. Entropy-regularized Optimal Transport adds a smoothing term:

$$\min_{\pi\in\Pi(\mu,\nu)} \langle c, \pi\rangle + \varepsilon\, \mathrm{KL}\big(\pi \,\|\, \mu\otimes\nu\big) \tag{3}$$

This produces a soft transport plan that is strongly convex and yields a unique solution.

It can be solved efficiently via the Sinkhorn algorithm. A brief route to the Sinkhorn form is to write the problem in the discrete setting, where $\pi \in \mathbb{R}_+^{n\times m}$ has prescribed marginals $\pi\mathbf{1} = a$ and $\pi^\top\mathbf{1} = b$ (with $a \in \Delta^n$ and $b \in \Delta^m$). Expanding the KL term (up to constants independent of π) gives an equivalent objective of the form

$$\min_{\pi\geq 0} \sum_{i,j}\Big(c_{ij}\pi_{ij} + \varepsilon\,\pi_{ij}\log\pi_{ij}\Big) \quad \text{s.t.} \quad \pi\mathbf{1} = a,\ \pi^\top\mathbf{1} = b. \tag{4}$$

Introducing Lagrange multipliers $\alpha \in \mathbb{R}^n$ and $\beta \in \mathbb{R}^m$ for the marginal constraints and taking first-order optimality conditions yields, entrywise,

$$c_{ij} + \varepsilon\big(\log\pi_{ij} + 1\big) - \alpha_i - \beta_j = 0, \tag{5}$$

so the optimizer must satisfy

$$\pi^\star_{ij} = \exp\Big(\tfrac{\alpha_i}{\varepsilon}\Big)\,\exp\Big(-\tfrac{c_{ij}}{\varepsilon}\Big)\,\exp\Big(\tfrac{\beta_j}{\varepsilon}\Big). \tag{6}$$

Defining $u_i := \exp(\alpha_i/\varepsilon)$, $v_j := \exp(\beta_j/\varepsilon)$, and $K_{ij} := \exp(-c_{ij}/\varepsilon)$ gives the scaling structure $\pi^\star = \mathrm{diag}(u)\,K\,\mathrm{diag}(v)$, where u and v are chosen so that the prescribed marginals are met. Computing these scalings is precisely an instance of matrix scaling / iterative proportional fitting: alternately enforcing the row and column sums amounts to alternating KL projections onto the affine sets defined by the two marginal constraints. Under mild conditions, these alternating projections contract in the Hilbert projective metric and converge to the unique pair of scalings, hence to the unique optimal plan.

Concretely, the Sinkhorn algorithm exploits this factorization and alternates row and column normalizations:

$$\pi^\star = \mathrm{diag}(u)\,K\,\mathrm{diag}(v), \qquad K_{ij} = \exp\big(-c_{ij}/\varepsilon\big), \tag{7}$$

with updates

$$u \leftarrow a \oslash (Kv), \qquad v \leftarrow b \oslash (K^\top u), \tag{8}$$

where a and b are the discrete marginals.

2.2 The Schrödinger Bridge Formulation

The Schrödinger Bridge is the dynamic analogue of entropy-regularized optimal transport, where the goal is not only to match endpoint distributions but to infer an entire stochastic transport process over time. In the discrete-time setting, consider a path $X_{0:N} = (X_0, X_1, \ldots, X_N)$ on a state space, and let P_{ref} be a reference path measure (typically Markovian) that encodes a preference for smooth or likely transitions. The Schrödinger Bridge problem seeks a path measure P that stays close to P_{ref} in KL divergence while matching prescribed endpoints:

$$\min_P\ \mathrm{KL}\big(P \,\|\, P_{\mathrm{ref}}\big) \qquad \text{s.t.} \qquad P_0 = \mu,\ \ P_N = \nu. \tag{9}$$

This yields not just a coupling between μ and ν, but a full sequence of intermediate marginals $\{P_n\}_{n=0}^N$ and a consistent joint law on trajectories.

A useful sanity check is the special case $N = 1$, where there is only a single transition $X_{0:1} = (X_0, X_1)$ and no intermediate marginals. In this case the feasible set consists of couplings $\pi = P_{0,1} \in \Pi(\mu,\nu)$, and the Schrödinger Bridge reduces to the static I-projection problem

$$\min_{\pi\in\Pi(\mu,\nu)}\ \mathrm{KL}\big(\pi \,\|\, R\big), \tag{10}$$

where $R := P_{\text{ref}}(X_0, X_1)$ is the two-point reference measure induced by P_{ref}. This recovers the usual entropy-regularized optimal transport objective as follows: if the reference is chosen as the Gibbs kernel

$$R_{ij} \;=\; \frac{1}{Z}\,\mu_i\,\nu_j\,\exp\!\big(-c_{ij}/\varepsilon\big) \qquad (Z>0 \text{ normalizes } R), \tag{11}$$

then

$$\mathrm{KL}(\pi\|R) = \mathrm{KL}(\pi\|\mu\otimes\nu) + \frac{1}{\varepsilon}\,\langle c,\pi\rangle + \text{const}, \tag{12}$$

so minimizing $\mathrm{KL}(\pi\|R)$ over $\Pi(\mu,\nu)$ is equivalent to minimizing

$$\langle c,\pi\rangle + \varepsilon\,\mathrm{KL}\big(\pi\,\|\,\mu\otimes\nu\big), \tag{13}$$

the standard entropic optimal transport problem.

For general N, the discrete Schrödinger Bridge can be solved efficiently by iterative scaling methods such as Iterative Proportional Fitting (IPF). IPF can be viewed as the dynamic analogue of the Sinkhorn algorithm: rather than scaling a matrix to match two marginals, it rescales a path measure to match the endpoint marginals. Conceptually, IPF is an alternating KL-projection scheme that iteratively projects the current path measure onto the affine constraint sets enforcing the correct initial and terminal marginals. Denote

$$\mathcal{C}_0 := \{P:\ P_0=\mu\}, \qquad \mathcal{C}_N := \{P:\ P_N=\nu\}. \tag{14}$$

Starting from $P^{(0)} = P_{\text{ref}}$, IPF alternates

$$Q^{(k)} = \arg\min_{Q\in\mathcal{C}_N}\ \mathrm{KL}\big(Q\,\|\,P^{(k)}\big), \qquad P^{(k+1)} = \arg\min_{P\in\mathcal{C}_0}\ \mathrm{KL}\big(P\,\|\,Q^{(k)}\big). \tag{15}$$

These KL projections have simple closed forms: enforcing $P_0=\mu$ amounts to multiplying the path weights by a factor depending only on x_0, and enforcing $P_N=\nu$ multiplies by a factor depending only on x_N. Under mild positivity conditions on P_{ref}, these alternating KL projections converge to the unique minimizer.

A related alternative is Iterative Markovian Fitting (IMF), which is often used when one wants to keep iterates explicitly Markov. It alternates between (i) a KL projection onto the set of Markov path measures and (ii) an endpoint-marginal correction step, which is a KL projection enforcing $P_0=\mu$ and $P_N=\nu$, and likewise converges under standard conditions.

2.3 Discrete Diffusion Models

In modern discrete generative modeling, standard diffusion operates over a finite state space of K categories (Austin et al., 2021). Let $x_0 \sim p(x_0) = \mu_0$ denote the categorical data distribution. We define a forward Markov process $p(x_t \mid x_{t-1})$ that gradually injects noise into the system. The objective of this process is to evolve the initial data distribution toward a known target distribution $p(x_T) = \mu_T$. The backward process $q_\phi(x_{t-1} \mid x_t)$ is subsequently trained to approximate the true reverse dynamics, enabling iterative sampling from μ_T back to μ_0:

$$p(x_{1:T} \mid x_0) = \prod_{t=1}^{T} p(x_t \mid x_{t-1}), \quad p_(x_0) = \mu_0, \quad p(x_T) = \mu_T \tag{16}$$

$$q_\phi(x_{0:T}) = q_\phi(x_T)\prod_{t=1}^{T} q_\phi(x_{t-1} \mid x_t), \quad q_\phi(x_T) = \mu_T \tag{17}$$

A common choice for the transition kernel is the uniform transition structure. In this case, the transition matrix at step t is defined as $P_t = (1-\varepsilon_t)\mathbf{I} + \frac{\varepsilon_t}{K}\mathbf{1}\mathbf{1}^\top$, which takes the form:

$$P_t^{\text{uniform}} = \begin{bmatrix} 1-\frac{K-1}{K}\varepsilon_t & \frac{\varepsilon_t}{K} & \cdots & \frac{\varepsilon_t}{K} \\ \frac{\varepsilon_t}{K} & 1-\frac{K-1}{K}\varepsilon_t & \cdots & \frac{\varepsilon_t}{K} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\varepsilon_t}{K} & \frac{\varepsilon_t}{K} & \cdots & 1-\frac{K-1}{K}\varepsilon_t \end{bmatrix} \tag{18}$$

By the property of Markov chains, the transition from the initial state x_0 to state x_t is given by the integrated transition matrix $P_{1:t} = P_1 P_2 \dots P_t$. Due to the algebraic properties of uniform transitions, this simplifies to:

$$P_{1:t} = (1-\bar{\varepsilon}_t)\mathbf{I} + \frac{\bar{\varepsilon}_t}{K}\mathbf{1}\mathbf{1}^\top, \quad \text{where} \quad \bar{\varepsilon}_t = 1-\prod_{i=1}^{t}(1-\varepsilon_i). \tag{19}$$

As $T \to \infty$ and $\bar{\varepsilon}_T \to 1$, the matrix $P_{1:T}$ converges to the rank-one matrix $\frac{1}{K}\mathbf{1}\mathbf{1}^\top$, ensuring that μ_T becomes uniform and independent of the initial distribution μ_0.

The backward generative process employs an x_0-parameterization, wherein a neural network predicts the categorical distribution $\hat{q}_\phi(x_0 \mid x_t)$. This prediction is used to compute the reverse transition via marginalization over the predicted state:

$$q_\phi(x_{t-1} \mid x_t) \propto \sum_{x_0} p(x_{t-1} \mid x_t, x_0)\hat{q}_\phi(x_0 \mid x_t). \tag{20}$$

Training proceeds by minimizing the Variational Lower Bound (VLB) combined with a supplementary cross-entropy loss to enforce accurate reconstruction of the original signal x_0:

$$\begin{aligned}
\mathcal{L} &= \mathcal{L}_{\text{vb}} + \lambda \mathcal{L}_{\text{ce}} \\
\mathcal{L}_{\text{vb}} &= \mathbb{E}_{p(x_0)}\Big[\underbrace{-\mathbb{E}_{p(x_1|x_0)}[\log q_\phi(x_0 \mid x_1)]}_{\text{Reconstruction}} + \underbrace{D_{\text{KL}}(p(x_T \mid x_0) \parallel q_\phi(x_T))}_{\text{Prior Matching}} \\
&\quad + \sum_{t=2}^{T} \underbrace{\mathbb{E}_{p(x_t|x_0)}[D_{\text{KL}}(p(x_{t-1} \mid x_t, x_0) \parallel q_\phi(x_{t-1} \mid x_t))]}_{\text{Denoising Matching}} \Big] \\
\mathcal{L}_{\text{ce}} &= \mathbb{E}_{p(x_0)}\mathbb{E}_{p(x_t|x_0)}[-\log q_\phi(x_0 \mid x_t)]
\end{aligned} \tag{21}$$

The posterior $p(x_{t-1} \mid x_t, x_0)$ is analytically tractable and computed via Bayes' rule:

$$p(x_{t-1} \mid x_t, x_0) = \frac{p(x_t \mid x_{t-1})p(x_{t-1} \mid x_0)}{p(x_t \mid x_0)} = \text{Cat}\left(x_{t-1}; p = \frac{(x_t P_t^\top) \odot (x_0 P_{1:t-1})}{x_0 P_{1:t} x_t^\top}\right). \tag{22}$$

During training, we sample $t \sim \mathcal{U}\{1, \ldots, T\}$ and optimize the total loss $\mathcal{L}$ with respect to the parameters ϕ.

3 Methodology: Bridging Diffusion and IPF

3.1 Connection to Iterative Proportional Fitting

In the standard diffusion paradigm, the target distribution $p_\theta(x_T)$ is typically fixed as an uninformative noise prior, such as the uniform distribution previously discussed. By generalizing this framework such that $p_\theta(x_T) = \mu_T$, the forward process $p_\theta(x_t \mid x_{t-1})$ is no longer pre-determined and becomes a trainable component of the system parametrized by θ. By reformulating the VLB, we observe that the objective corresponds to the KL minimization of the trajectory distributions. For the backward process, this is expressed as:

$$\arg\max_\phi \mathbb{E}_{p_\theta(x_{0:T})}\left[\log \frac{q_\phi(x_{0:T})}{p_\theta(x_{1:T} \mid x_0)}\right] = \arg\min_\phi \text{KL}(p_\theta(x_{0:T}) \parallel q_\phi(x_{0:T})). \tag{23}$$

Symmetrically, optimizing the forward generative process involves the reverse objective:

$$\arg\min_\theta \text{KL}(q_\phi(x_{0:T}) \parallel p_\theta(x_{0:T})). \tag{24}$$

This alternating optimization scheme—where one process is held stationary while the other is refined to match the joint trajectory—mathematically resembles the IPF algorithm. Specifically, this version utilizes M-projections instead of the traditional information projections (I-projections). In the context of generative modeling, this perspective frames diffusion as a sequence of Schrödinger Bridge problems, wherein p_θ and q_ϕ iteratively refine their respective transition densities to satisfy the marginal constraints at $t = 0$ and $t = T$.

3.2 The "Lazy" Mixture Transition

A central challenge in discrete SB is the necessity of ensuring a smooth transition between the boundary distributions. In the absence of explicit regularization, the model tends to learn a trivial mapping that jumps from the source to the target distribution in a single step, failing to produce a meaningful sequence of intermediate states. To enforce a gradual, step-wise evolution, we define a reference process that strongly favors the identity mapping, effectively implementing a "lazy" random walk mixture:

$$\text{Forward: } x_{t+1} \sim \tilde{p}_\theta(\cdot \mid x_t), \qquad \tilde{p}_\theta(\cdot \mid x_t) = (1-\varepsilon)\delta_{x_t} + \varepsilon p_\theta(\cdot \mid x_t) \tag{25}$$

$$\text{Backward: } x_{t-1} \sim \tilde{q}_\phi(\cdot \mid x_t), \qquad \tilde{q}_\phi(\cdot \mid x_t) = (1-\varepsilon)\delta_{x_t} + \varepsilon q_\phi(\cdot \mid x_t) \tag{26}$$

In this framework, the term $(1 - \varepsilon)$ acts as a regularization weight that encourages the state to persist across timesteps, while ε governs the updates proposed by the neural model. This mixing rule ensures that the learned bridge remains locally smooth and temporally consistent, compelling the network to discover a continuous trajectory that satisfies the endpoint constraints.

3.3 Neural IPF: Path-Space Factorization and I-Projections

In an ideal Schrödinger Bridge formulation, learning can proceed via IPF, which alternates between I-projections in trajectory space. Rather than matching distributions only at the endpoints, the goal is to align full path measures over sequences $x_{0:T}$.

Because the bridge dynamics are Markovian, both forward and backward path measures admit equivalent factorizations. In particular,

$$P_\theta(x_{0:T}) = P_\theta(x_0) \prod_{t=1}^{T} p_\theta(x_t \mid x_{t-1}) = P_\theta(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t),$$

$$Q_\phi(x_{0:T}) = Q_\phi(x_T) \prod_{t=1}^{T} q_\phi(x_{t-1} \mid x_t),$$

where the first equality expresses the usual forward Markov decomposition, while the second highlights the time-reversed parameterization that is natural for IPF updates.

An I-projection corresponds to minimizing a KL divergence over path space while enforcing a marginal constraint. Assuming that the terminal marginal of the auxiliary process is fixed, $Q_\phi(x_T) = \mu_T(x_T)$, the forward projection step minimizes $\mathrm{KL}(P_\theta \| Q_\phi)$. Using the chain rule for KL divergences between Markov measures, this objective decomposes into a terminal marginal term and a sum of conditional KL terms:

$$\mathrm{KL}(P_\theta \| Q_\phi) = \mathbb{E}_{x_{0:T} \sim P_\theta}\left[\log \frac{P_\theta(x_{0:T})}{Q_\phi(x_{0:T})}\right] = \overbrace{\mathrm{KL}\big(P_\theta(x_T) \,\|\, \mu_T\big)}^{\text{terminal marginal penalty}} + \sum_{t=1}^{T} \mathbb{E}_{x_t \sim P_\theta} \mathrm{KL}(P_\theta(x_{t-1} \mid x_t) \,\|\, q_\phi(x_{t-1} \mid x_t)) \,. \tag{27}$$

This decomposition provides an important interpretation: ideal IPF simultaneously enforces the endpoint constraint $P_\theta(x_T) = \mu_T$ and aligns each reverse-time transition via local KL projections. In other words, the global path-space optimization reduces to a sequence of conditional matching problems.

However, performing this I-projection exactly is infeasible in our neural setting. The quantities required by the objective — namely the true reverse conditionals $P_\theta(x_{t-1} \mid x_t)$ and the induced terminal marginal $P_\theta(x_T)$ — depend on integrating over exponentially many trajectories. Unlike classical Schrödinger Bridge solvers where the reference process admits analytic forward–backward recursions, neural stochastic processes do not provide tractable closed-form marginals or posteriors. Consequently, computing the exact KL terms or sampling from the required conditionals becomes computationally intractable, motivating the M-projection approximation introduced in the next part.

3.4 Practical Training via M-Projections

Since exact I-projections require access to intractable reverse-time conditionals of the neural path measure, we replace them with M-projection updates. Rather than projecting onto the constraint set by minimizing $\mathrm{KL}(P_\theta \| Q_\phi)$ with respect to the full path distribution, we fix the current bridge process P_θ and update only the local transition model q_ϕ.

Under this setting, terms in the KL objective that do not depend on ϕ collapse into a constant C_θ. Using the factorization of $Q_\phi(x_{0:T})$, the forward KL reduces to

$$\mathrm{KL}(P_\theta \| Q_\phi) = C_\theta - \sum_{t=1}^{T} \mathbb{E}_{x_{0:T} \sim P_\theta}\big[\log q_\phi(x_{t-1} \mid x_t)\big], \tag{28}$$

which is equivalent to maximizing the log-likelihood of reverse-time transitions sampled from the current bridge dynamics.

This observation provides a practical training rule: instead of matching unknown path-space conditionals, we perform local Maximum Likelihood Estimation (MLE) on pairs (x_{t-1}, x_t) generated by P_θ. The resulting update is computationally tractable, requiring only sampling and standard cross-entropy optimization, and therefore scales naturally to neural parameterizations.

Conceptually, this replaces the global I-projection with a sequence of local M-projections that fit conditional transitions while treating trajectory samples as empirical targets. While efficient, this approximation introduces a key limitation: because the optimization no longer contains the explicit marginal penalty $\mathrm{KL}(P_\theta(x_T)\|\mu_T)$, the endpoint distributions are not guaranteed to match the desired Schrödinger boundary conditions. As a result, additional mechanisms are required to restore marginal alignment, which we address via adversarial matching in the following part.

3.5 Restoring Bounds with Adversarial Marginal Matching

Although M-projection updates provide a tractable training rule, they remove the explicit endpoint penalties that arise in the ideal I-projection objective. In particular, the marginal KL terms $\mathrm{KL}(P_\theta(x_T)\|\mu_T)$ and $\mathrm{KL}(Q_\phi(x_0)\|\mu_0)$ are no longer directly optimized, which can lead to drift between the learned bridge endpoints and the desired Schrödinger boundary distributions.

To restore these constraints, we introduce an adversarial marginal matching strategy. Instead of computing marginal KL divergences explicitly — which would require access to intractable densities — we train discriminator networks that provide a learned divergence signal between generated endpoint samples and the target marginals. Specifically, we employ two hinge-loss critics:

- D_T, which distinguishes samples from the forward bridge endpoint $P_\theta(x_T)$ and the true target distribution μ_T,
- D_0, which distinguishes samples from the backward bridge endpoint $Q_\phi(x_0)$ and the source distribution μ_0.

From a theoretical perspective, these critics act as variational estimators of the missing marginal divergences, providing gradients that encourage endpoint alignment while leaving intermediate transition learning governed by the M-projection likelihood objective. This hybrid optimization can be interpreted as replacing explicit marginal KL terms with learned f-divergence surrogates implemented via adversarial training (Nowozin et al., 2016).

Our final learning scheme therefore combines two complementary signals: (i) local transition fitting through M-projection MLE updates, which promote smooth and stable trajectory dynamics, and (ii) adversarial endpoint matching, which enforces the Schrödinger boundary conditions at $t = 0$ and $t = T$. Together, these components approximate the behavior of ideal path-space projections while remaining computationally feasible for neural discrete diffusion models.

4 Experiments and Results

4.1 Experimental Setup

We evaluate our discrete Schrödinger Bridge framework on a domain translation task between MNIST handwritten digits and EMNIST handwritten letters. Images are treated as discrete token sequences by uniformly quantizing grayscale intensities into $K = 16$ categorical states. This allows both forward and backward bridge dynamics to operate in a finite state space consistent with the discrete diffusion formulation described in previous sections.

Both bridge processes P_θ and Q_ϕ are parameterized using a shared time-conditioned U-Net architecture operating on one-hot token embeddings. The networks receive either hard token inputs or soft one-hot representations depending on the training stage. Endpoint critics D_T and D_0 are convolutional discriminators trained with hinge adversarial losses and R1 gradient regularization.

Training follows an alternating Neural IPF procedure for 2000 outer iterations with a lazy transition kernel $\tilde{p}(\cdot|x) = (1 - \varepsilon)\delta_x + \varepsilon p_\theta(\cdot|x)$ using $\varepsilon = 0.1$, which encourages gradual semantic transformations along the bridge.

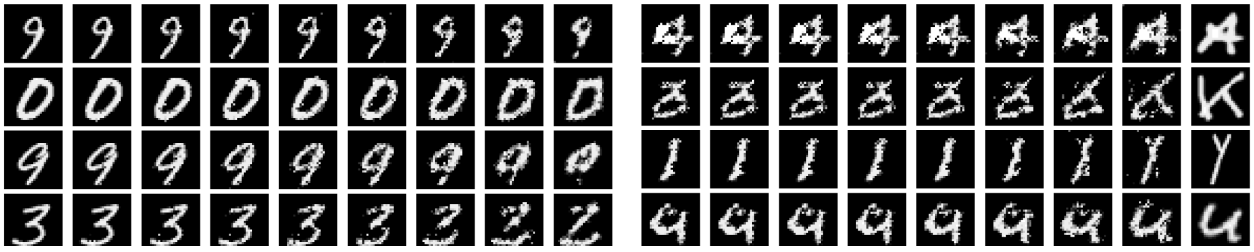
4.2 Implementation Details

Each training iteration alternates between forward and backward M-projection updates. First, trajectories are sampled from the current forward bridge P_θ using hard discrete sampling. These trajectories provide supervision pairs (x_{t-1}, x_t) used to fit the reverse model Q_ϕ via local maximum likelihood estimation. The same procedure is then performed in reverse to update P_θ using samples from Q_ϕ.

To enable adversarial endpoint matching, we generate differentiable rollouts using a straight-through Gumbel-Softmax relaxation. These soft trajectories allow gradients from the endpoint critics to propagate through multiple bridge steps without requiring explicit likelihood evaluation of marginal distributions. The Gumbel temperature is annealed during training to transition from smooth exploratory dynamics toward sharper discrete predictions.

Discriminators are trained with hinge losses and an additional R1 gradient penalty to stabilize adversarial optimization. Because trajectory sampling across $T = 80$ steps is memory intensive, we employ truncated backpropagation over time ("chunked" updates) to control GPU usage while preserving training stability.

4.3 Bridge Dynamics and Trajectory Analysis



(a) Forward P_θ (MNIST $\rightarrow$ EMNIST) (b) Backward Q_ϕ (EMNIST $\rightarrow$ MNIST)

Figure 1: Bridge samples across discrete time steps. Forward P_θ morphs digits into letters; backward Q_ϕ reverses the path. Endpoints match target marginals via adversarial matching, while intermediate states smoothly interpolate.

Figure 1 visualizes intermediate bridge states. Unlike standard diffusion processes that collapse rapidly toward noise, our lazy discrete bridge exhibits gradual semantic changes over time. Early steps preserve digit structure while later steps progressively reshape strokes toward letter-like patterns. The backward process exhibits symmetric behavior, indicating that the learned stochastic dynamics approximate a consistent transport path between the two domains.

Qualitative translations in Figure 2 demonstrate that the learned bridge performs coherent domain adaptation while maintaining structural consistency. Because the process evolves through a sequence of intermediate distributions rather than a direct mapping, the model avoids the abrupt semantic shifts often observed in deterministic translation methods. These results highlight the advantage of Schrödinger Bridge dynamics for structured discrete transport tasks.

4.4 Training Algorithm

Algorithm 1 summarizes our practical Neural IPF training procedure. Each outer iteration alternates between (i) M-projection updates that fit local transitions using hard sampled trajectory pairs, and (ii) adversarial marginal matching at the endpoints using differentiable straight-through (ST) Gumbel-Softmax rollouts. Discriminators are trained with hinge losses and an R1 gradient penalty for stability.



(a) MNIST → EMNIST Translation (b) EMNIST → MNIST Translation

Figure 2: Qualitative translation results. Top rows indicate original inputs, while bottom rows display the transformed domain outputs. High-fidelity samples successfully adapt to the new domain's semantic structures.

Algorithm 1: Neural IPF with M-Projections and Adversarial Endpoint Matching

Input: Source samples $x_0 \sim \mu_0$ (MNIST), target samples $x_T \sim \mu_T$ (EMNIST)
Input: Bridge models P_θ, Q_ϕ, critics D_T, D_0, steps T, lazy rate ε
Input: Gumbel temperature schedule $\tau(it)$, GAN weight $\lambda_{\text{gan}}(it)$
for $it = 1, 2, \dots$ do

// (1) M-projection update for Q_ϕ using hard pairs from P_θ
Sample $x_0 \sim \mu_0$
Generate a hard forward trajectory $\{x_t\}_{t=0}^{T}$ via the lazy kernel $\tilde{p}_\theta(\cdot|x_t) = (1-\varepsilon)\delta_{x_t} + \varepsilon p_\theta(\cdot|x_t)$
for $t = 0, \dots, T-1$ do
Update Q_ϕ by minimizing the local NLL $-\log \tilde{q}_\phi(x_t \mid x_{t+1})$

// Adversarial endpoint matching for Q_ϕ: enforce $Q_\phi(x_0) \approx \mu_0$
Sample $x_T \sim \mu_T$
Roll out a differentiable backward trajectory $\tilde{x}_0$ from x_T using ST Gumbel-Softmax with temperature $\tau(it)$
Update Q_ϕ with generator hinge loss $L_{\text{adv}}^{Q} = -\mathbb{E}[D_0(\tilde{x}_0)]$

// (2) Critic updates (hinge + R1)
Sample real $x_T \sim \mu_T$ and fake $\tilde{x}_T \leftarrow P_\theta(x_0)$ via ST rollout
Update D_T using $L_{D_T} = \mathbb{E}[\max(0, 1 - D_T(x_T))] + \mathbb{E}[\max(0, 1 + D_T(\tilde{x}_T))] + \gamma R_1$
Sample real $x_0 \sim \mu_0$ and fake $\tilde{x}_0 \leftarrow Q_\phi(x_T)$ via ST rollout
Update D_0 using the analogous hinge + R1 objective

// (3) M-projection update for P_θ using hard pairs from Q_ϕ
Sample $x_T \sim \mu_T$
Generate a hard backward trajectory $\{x_t\}_{t=0}^{T}$ via the lazy kernel $\tilde{q}_\phi(\cdot|x_{t+1}) = (1-\varepsilon)\delta_{x_{t+1}} + \varepsilon q_\phi(\cdot|x_{t+1})$
for $t = 0, \dots, T-1$ do
Update P_θ by minimizing the local NLL $-\log \tilde{p}_\theta(x_{t+1} \mid x_t)$

// Adversarial endpoint matching for P_θ: enforce $P_\theta(x_T) \approx \mu_T$
Sample $x_0 \sim \mu_0$
Roll out a differentiable forward trajectory $\tilde{x}_T$ from x_0 using ST Gumbel-Softmax
Update P_θ with generator hinge loss $L_{\text{adv}}^{P} = -\mathbb{E}[D_T(\tilde{x}_T)]$

5 Conclusion and Future Work

We introduced a practical framework for learning Discrete Schrödinger Bridges by combining IPF-inspired alternating updates with tractable M-projection training and adversarial marginal matching. By replacing intractable path-space I-projections with local maximum-likelihood updates, our method enables scalable training of stochastic transport processes in discrete domains while retaining meaningful intermediate trajectory structure. The addition of adversarial endpoint critics restores the boundary constraints that are lost under pure M-projection optimization, allowing the learned bridge to approximate the behavior of an ideal entropy-regularized transport process.

Empirically, our experiments demonstrate that the resulting discrete bridge produces smooth semantic transitions between MNIST digits and EMNIST letters, avoiding the abrupt transformations typical of deterministic translation models. These results highlight the potential of Schrödinger Bridge formulations as a principled alternative to pure-noise diffusion and cycle-consistent adversarial approaches for structured domain transfer.

Several directions remain for future work. First, we plan to replace the current hinge-loss critics with a KL-based f-GAN (Nowozin et al., 2016) objective, which would more directly approximate the marginal divergence terms $\mathrm{KL}(P_\theta(x_T)\|\mu_T)$ and $\mathrm{KL}(Q_\phi(x_0)\|\mu_0)$ and therefore bring the practical training procedure closer to the theoretical Schrödinger Bridge objective. Second, scaling to higher-resolution imagery may be achieved by performing transport in a discrete latent space obtained through vector-quantized tokenization (e.g., VQGAN (Esser et al., 2021)), allowing the bridge dynamics to operate over semantically meaningful discrete codes rather than raw pixels.

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